Effat University

Lab 3: Requirement Engineering

SPRING 2026 – CS3151

ENGINEERING COLLEGE – COMPUTER SCIENCE DEPARTMENT

INSTRUCTORS NAME: Dr. Passent Elkafrawy		
COURSE NAME: Software Engineering	COURSE NUMBER: CS3151	SECTION: 1-2-3

Transforming User Stories to Requirements

Movie Shop Case Study

Learning Objectives

By completing this lab, you will be able to:

1. Identify and extract the three key components of a user story: Role (Actor), Feature (Function), and Benefit (Object)
2. Transform informal user stories into formal, SMART requirements
3. Apply the SMART criteria (Specific, Measurable, Achievable, Relevant, Time-bound) to requirements
4. Recognize how a single user story can expand into multiple detailed requirements

---

Part 1: Understanding the Movie Shop Domain

The Movie Shop is a retail business that sells and rents both physical and digital videos. The system must support multiple actors (customers, members, sales clerks, and managers) with various functionalities. Before transforming user stories, familiarize yourself with the key entities and actors in this domain:

Key Actors:

- Customer (Non-member): Can purchase/rent physical videos, view reviews, submit reviews
- Member: Pays annual fee, gets 10% discount, can make reservations, access digital videos
- Sales Clerk: Manages transactions, customer information, video inventory
- Shop Manager: Generates reports, oversees operations
- System: Handles video inventory, rental tracking, overdue notices

Key Entities:

- Physical Videos (copies with rental/sale tracking)
- Digital Videos (no physical copies)
- Customers/Members (with personal information)

- Rentals (with due dates and overdue tracking)
- Reservations (up to 5 per member)
- Reviews (with ratings 1-10, up to 100 words, optional anonymity

---

Part 2: User Story Template and Structure

A well-formed user story follows this structure:

```
As a [ROLE/ACTOR],
I want to [FEATURE/FUNCTION],
so that [BENEFIT/OBJECT].
```

Key Components:

Component	Definition	Example
Role (Actor)	Who is performing the action?	Customer, Member, Sales Clerk, Manager
Feature (Function)	What action or capability is needed?	Add to cart, make reservation, generate report
Benefit (Object)	Why is this needed? What value does it provide?	Purchase multiple items at once, ensure availability, track performance

---

Part 3: User Stories from Movie Shop Examples

Below are user stories extracted from the Movie Shop case study. Study each one and identify the Role, Feature, and Benefit.

User Story 1: Member Discount on Purchases

```
As a member,
I want to receive a 10% discount on video sales and rentals,
so that I can save money on my purchases and feel valued for my membership.
```

Role (Actor): Member
Feature (Function): Receive discount on sales and rentals
Benefit (Object): Save money, feel valued

User Story 2: Make Reservations Online

```
As a member,
I want to make reservations for physical video rentals via the Web,
so that I can secure videos I want without visiting the shop.
```

Role (Actor): Member
Feature (Function): Make reservations via Web
Benefit (Object): Secure videos in advance, convenience

User Story 3: Submit Video Reviews

```
As a customer,
I want to submit a review and rating for videos I have purchased or rented,
so that I can share my opinion and help other customers make purchasing decisions.
```

Role (Actor): Customer
Feature (Function): Submit review and rating
Benefit (Object): Share opinions, help other customers

User Story 4: Anonymous Reviews

```
As a customer,
I want to choose whether my name appears with my review,
so that I can protect my privacy while still sharing my feedback.
```

Role (Actor): Customer
Feature (Function): Choose whether my name appears with my review
Benefit (Object): Protect my privacy while still sharing feedback

---

User Story 5: Track Overdue Rentals

```
As a shop manager,
I want the system to automatically track overdue physical video rentals,
so that I can ensure timely collection and maintain inventory control.
```

Role (Actor): Shop manager
Feature (Function): Have the system automatically track overdue physical rentals
Benefit (Object): Ensure timely collection and keep inventory under control

---

User Story 6: Manage Customer Information

```
As a sales clerk,
I want to enter and update customer information (name, address, phone, age, sex, email),
so that I can maintain accurate customer records for sales and communication.
```

Role (Actor): Sales clerk
Feature (Function): Enter and update customer information (name, address, phone, age, sex, email)
Benefit (Object): Keep accurate customer records for sales and communication

---

User Story 7: Look Up Customer Status

```
As a sales clerk,
I want to quickly look up whether a customer is a member,
so that I can apply appropriate discounts during transactions.
```

Role (Actor): Sales Clerk
Feature (Function): Look up customer membership status
Benefit (Object): Apply correct discounts, ensure accuracy

User Story 8: Manage Video Inventory

```
As a sales clerk,
I want to enter and update video information (title, director, genre, price, etc.),
so that I can maintain an accurate and searchable video catalog.
```

Role (Actor): Manage video inventory
Feature (Function): Sales clerk
Benefit (Object): Enter and update video information (title, director, genre, price)

User Story 9: Generate Sales Reports

```
As a shop manager,
I want to generate reports on video sales and rentals,
so that I can analyze business performance and make informed decisions.
```

Role (Actor): Generate sales reports
Feature (Function): Shop manager
Benefit (Object): Generate reports on video sales and rentals

---

User Story 10: Send Overdue Notices

```
As the system,
I want to automatically send email notices to customers with overdue physical video rentals,
so that customers are reminded to return videos and the shop can recover inventory.
```

Role (Actor): System
Feature (Function): Automatically email notices to customers with overdue physical rentals
Benefit (Object): Remind customers to return videos and let the shop recover inventory

---

Part 5: Exercises

Exercise 1: Extract Components from User Stories

For each user story below, identify and write down the Role (Actor), Feature (Function), and Benefit (Object):

User Story A:

```
As a customer,
I want to purchase digital videos online,
so that I can watch movies immediately without leaving home.
```

Your Answer:

- Role: Customer
- Feature: Purchase digital videos online
- Benefit: Watch movies immediately without leaving home

User Story B:

```
As a sales clerk,
I want to process the return of rented physical videos,
so that I can update inventory and check for damages.
```

Your Answer:

- Role: Sales clerk
- Feature: Process the return of rented physical videos
- Benefit: Update inventory and check for damages

User Story C:

```
As a member,
```

```
I want to change my personal information online using my membership number and password,
```

```
so that I can keep my account up to date.
```

Your Answer:

- Role: Member
- Feature: Change personal information online using membership number and password
- Benefit: Keep the account up to date

Part 4: Transformation Examples

Example 1: Member Discount Transformation

Original User Story:

```
As a member,
```

```
I want to receive a 10% discount on video sales and rentals,
```

```
so that I can save money on my purchases and feel valued for my membership.
```

Extracted Components:

- Role: Member
- Feature: Receive discount on sales and rentals
- Benefit: Save money, feel valued

Transformed into SMART Requirements:

Req ID	Requirement	SMART Criteria
REQ-1	The system must apply a 10% discount to all video sales for members.	S: Specific discount rate (10%); M: Discount amount calculated; A: Achievable with pricing logic; R: Supports business model; T: Applied at point of sale
	The system must apply a 10% discount to all video sales for members at point of scale	
REQ-2	The system must clearly display the discount amount during checkout.	S: Display discount; M: Discount shown in transaction; A: UI implementation; R: Transparency; T: At checkout
	The systems must display the discount during checkout in the transaction details page.	
REQ-3	The system must verify membership status before applying discounts.	S: Check membership; M: Status confirmed; A: Database lookup; R: Accuracy; T: Before discount applied
	The system must verify membership status by checking the membership database before applying the membership discount	

Example 2: Make Reservations Online Transformation

Original User Story:

```
As a member,
I want to make reservations for physical video rentals via the Web,
so that I can secure videos I want without visiting the shop.
```

Extracted Components:

- Role: Member
- Feature: Make reservations via Web
- Benefit: Secure videos in advance, convenience

Transformed into SMART Requirements:

Req ID	Requirement	SMART Criteria
REQ-5	Members must be able to reserve physical videos through the Web interface.	S: Web-based reservation; M: Reservation created; A: Web development; R: Convenience; T: Available 24/7
REQ-6	A member can reserve a maximum of 5 physical videos at any one time.	S: Maximum 5 reservations; M: Count enforced; A: Business rule; R: Inventory management; T: Per reservation attempt
REQ-7	The system must confirm reservations and display confirmation details to the member.	S: Confirmation provided; M: Confirmation displayed; A: UI implementation; R: User confirmation; T: Immediately after reservation
REQ-8	The system must notify members when reserved videos become available for pickup.	S: Notification sent; M: Email/notification delivered; A: Notification system; R: Customer engagement; T: When video available
REQ-9	Members must be able to make reservations by phone or in person in addition to the Web.	S: Multiple channels; M: Reservations accepted via phone/in-person; A: Staff training; R: Accessibility; T: During business hours

---

Example 3: Submit Video Reviews Transformation

Original User Story:

```
As a customer,
I want to submit a review and rating for videos I have purchased or rented,
so that I can share my opinion and help other customers make purchasing decisions.
```

Extracted Components:

- Role: Customer
- Feature: Submit review and rating
- Benefit: Share opinions, help other customers

Transformed into SMART Requirements:

Req ID	Requirement	SMART Criteria
REQ-10	The system must allow customers to submit reviews for videos they have purchased or rented.	S: Review submission; M: Review stored; A: Database storage; R: Customer engagement; T: After purchase/rental
REQ-11	Reviews must include a rating from 1 (lowest) to 10 (highest).	S: Rating scale 1-10; M: Rating value validated; A: Input validation; R: Standardized feedback; T: Required field
REQ-12	Reviews must be limited to a maximum of 100 words.	S: 100-word limit; M: Word count enforced; A: Text validation; R: Concise feedback; T: Before submission
REQ-13	The system must display reviews to other customers when browsing videos.	S: Display reviews; M: Reviews visible; A: UI implementation; R: Inform purchasing decisions; T: On video detail page
REQ-14	Customers must be able to choose whether their name appears with their review.	S: Anonymity option; M: Name display toggled; A: User preference; R: Privacy protection; T: At submission

---

Example 4: Track Overdue Rentals Transformation

Original User Story:

```
As a shop manager,
I want the system to automatically track overdue physical video rentals,
so that I can ensure timely collection and maintain inventory control.
```

Extracted Components:

- Role: Shop Manager
- Feature: Track overdue rentals automatically
- Benefit: Ensure timely collection, maintain inventory

Transformed into SMART Requirements:

Req ID	Requirement	SMART Criteria
REQ-15	The system must automatically identify physical video rentals that are past their due date.	S: Identify overdue rentals; M: Overdue status flagged; A: Date comparison logic; R: Inventory management; T: Daily check

Req ID	Requirement	SMART Criteria
REQ-16	The system must maintain a list of all overdue rentals accessible to the shop manager.	S: Overdue list available; M: List generated; A: Database query; R: Management visibility; T: Real-time
REQ-17	The system must send email notices to customers with overdue rentals.	S: Email notification; M: Email delivered; A: Email system; R: Customer reminder; T: When overdue
REQ-18	The system must allow the shop manager to generate overdue rental reports.	S: Report generation; M: Report created; A: Reporting module; R: Analysis and tracking; T: On demand
REQ-19	The system must track which copy of a physical video is rented and when it is due back.	S: Track copy and due date; M: Data recorded; A: Database storage; R: Inventory accuracy; T: At rental

---

Exercise 2: Transform User Stories to SMART Requirements

Transform the following user story into at least 5 SMART requirements. Use the format provided:

User Story:

```
As a member,
I want to track my rental history,
so that I can remember which videos I have rented and when.
```

Your Transformed Requirements:

Req ID	Requirement	SMART Criteria
REQ-1	The system must record the video title, rental date, due date and return date of every rental in the member's rental history at the moment the rental is created.	S: four named fields; M: one history record exists per rental; A: simple database write; R: the history can only be shown if it is recorded; T: at the moment of rental
REQ-2	A logged-in member must be able to view their own rental history through the Web interface.	S: own history only, via the Web; M: page opens and shows the member's records; A: standard authenticated page; R: the core of the story; T: available 24/7

Req ID	Requirement	SMART Criteria
	The rental history must list rentals sorted by rental date, newest first, showing title, rental date and return date.	S: fixed sort order and columns; M: order and columns can be checked on screen; A: simple query; R: helps the member remember what and when; T: every time the page loads
REQ-3	The rental history page must load within 3 seconds for a member with up to 500 records.	S: 3-second limit, 500 records; M: measured page load time; A: indexed query; R: usability; T: on each page load
	The member must be able to filter the history by date range or by video title.	S: two filter types; M: only matching records are shown; A: standard search; R: finds an old rental quickly; T: results appear within 2 seconds of applying the filter
	The system must keep each rental record in the history for at least 3 years after the return date.	S: 3-year retention; M: records older than 3 years are the only ones removable; A: storage is cheap; R: the member can look far back; T: 3 years from the return date
	The rental history must list rentals sorted by rental date, newest first, showing title, rental date and return date.	S: fixed sort order and columns; M: order and columns can be checked on screen; A: simple query; R: helps the member remember what and when; T: every time the page loads

---

Exercise 3: Identify Missing Requirements

The following user story has been transformed into requirements, but some important requirements are missing. Identify what is missing and add 2-3 additional requirements:

Original User Story:

```
As a sales clerk,
I want to sell physical videos to customers,
so that I can process transactions and maintain inventory.
```

Existing Requirements:

- REQ-1: The system must allow sales clerks to select a video for sale.
- REQ-2: The system must calculate the sale price for the video.
- REQ-3: The system must process payment from the customer.

What is missing? an availability check before the sale, an inventory update after the sale, a receipt or transaction record, the member discount, and what happens when payment fails.

Your Additional Requirements:

Req ID	Requirement
REQ-4	The system must check that at least one copy of the selected video is in stock before the sale is confirmed, and refuse the sale if none is left.
REQ-5	The system must reduce the stock count of the sold video by the quantity sold.
REQ-6	The system must record each sale (video, price, clerk, date and time, payment method) and give the customer a receipt.
	If payment fails, the system must cancel the sale, leave stock unchanged and let the clerk retry or choose another payment method.

Exercise 4: Create Your Own User Stories (Project Based)

Based on the Movie Shop case study, create new user stories for your project. For each story, ensure it follows the proper format and includes clear Role, Feature, and Benefit.

User Story 1:

```
As a Teacher,
```

```
I want to mark attendance for my class on my laptop
```

```
so that I save class time and avoid paper errors.
```

User Story 2:

```
As a parent,
```

```
I want toreceive an instant notification when my child is marked absent,
```

```
so that receive an instant notification when my child is marked absent.
```

User Story 3:

```
As a school administrator,
```

```
I want to timetable builder that warns me about room and teacher conflicts,
```

```
so that I can schedule classes without clashes.
```

Effat University

Lab 3: Requirement Engineering

SPRING 2026 – CS3151

Part 4: Group Project

Exercise 4. 1: Generate all user stories of your project

ID	As a...	I want to...	So that...
US-01	**Principal**	**see a dashboard of attendance rates and grade averages by class**	**I can monitor school performance from one place**
US-02	**Principal**	**see which teachers have taken attendance and entered grades today**	**I can follow up without chasing each teacher**
US-03	**Principal**	**receive an automatic weekly performance report**	**I can make decisions without asking staff for data**
US-04	**Teacher**	**record attendance digitally**	**I save class time and avoid paperwork errors**
US-05	**Teacher**	**see automatic grade calculations and alerts**	**I can identify struggling students early**
US-06	**Teacher**	**post assignments and track who has submitted**	**I know which students need a reminder**

US-07	Teacher	message a student's parents from the system	I can inform them quickly about problems
US-08	Student	view my timetable	I know where to be and when
US-09	Student	submit assignments and take quizzes online	I can finish my work in one place
US-10	Student	view my grades and attendance	I can follow my own progress
US-11	Parent	receive an instant notification when my child is absent	I can follow up quickly
US-12	Parent	pay school fees online	I avoid visiting the school and never miss a deadline
US-13	Parent	see my child's grades and attendance in real time	I know how my child is doing without calling the school

US-14	Parent	use the app in Arabic or English	I can understand everything comfortably
US-15	Parent	message my child's teacher	I can ask questions and explain absences
US-16	Administrator	enroll new students and manage their records	the school's records are complete and accurate
US-17	Administrator	use a timetable tool that detects conflicts	I can schedule classes without clashes
US-18	Administrator	generate attendance and performance reports automatically	I support decisions and reduce manual work
US-19	Administrator	track fee payments and outstanding balances	I can follow up on unpaid fees

US-20	IT support	create, disable and assign roles to user accounts	only authorized people see sensitive data
US-21	IT support	back up the data automatically and restore it when needed	the school never loses its records
US-22	System	send parents a reminder before each fee due date	fees are paid on time
US-23	System	log every access to student records	misuse can be detected and investigated

Exercise 4.2: Transform to requirements

Req ID	Story	Requirement	S / M / T check
FREQ-01	**US-01**	**The system shall display a dashboard showing attendance rate and grade average for each class of the current term.**	**S: two metrics per class; M: values equal the stored records; T: loads within 3 seconds**
FREQ-02	**US-01**	**The system shall restrict the dashboard to the principal and administrators.**	**S: two allowed roles; M: other roles receive "access denied"; T: checked on every request**
FREQ-03	**US-02**	**The system shall show the principal, for each teacher and each period of the day, whether attendance has been recorded.**	**S: status per teacher per period; M: "Done" or "Not done"; T: updated within 1 minute of recording**
FREQ-04	**US-03**	**The system shall email a weekly attendance and performance report to the principal.**	**S: weekly report with attendance and grade averages; M: email delivered; T: every Sunday at 7:00 AM (Riyadh time)**

FREQ-05	US-04	The system shall let a teacher mark every student of a class as present, absent or late on one screen.	S: three statuses, one screen; M: all students listed and saved; T: a class of 30 in under 1 minute
FREQ-06	US-04	The system shall make saved attendance visible to administrators, parents and students.	S: three viewer groups; M: record visible to them; T: within 1 minute of saving
FREQ-07	US-05	The system shall calculate a student's subject average automatically each time a grade is entered.	S: weighted average per subject; M: result equals the manual calculation; T: within 2 seconds
FREQ-08	US-05	The system shall alert the teacher when a student's average drops by 10 percentage points or more between two consecutive assessments.	S: 10-point drop rule; M: alert appears for every student meeting the rule; T: when the grade is saved

FREQ-09	US-06	The system shall let a teacher post an assignment with title, description, due date and one attachment of up to 20 MB.	S: four fields, 20 MB limit; M: assignment appears to the class; T: within 1 minute of posting
FREQ-10	US-06	The system shall show the teacher, for each assignment, which students have and have not submitted.	S: two lists per assignment; M: counts match submissions; T: updated within 1 minute of each submission
FREQ-11	US-07	The system shall let a teacher send a message to the parents of one student or of a whole class.	S: two recipient options; M: message appears in each parent's inbox; T: delivered within 1 minute
FREQ-12	US-08	The system shall show a student a weekly timetable with subject, teacher, room and time for every period.	S: four fields per period; M: matches the published timetable; T: available 24/7

FREQ-13	US-09	The system shall let a student submit an assignment file (PDF, Word or image, up to 20 MB) and show a confirmation.	S: three file types, 20 MB limit; M: confirmation with time stamp; T: within 5 seconds of upload
FREQ-14	US-09	The system shall reject submissions after the due date unless the teacher grants an extension.	S: rule with one exception; M: late upload blocked; T: applied from the due date and time
FREQ-15	US-09	The system shall run timed quizzes, submit them automatically when time ends and grade multiple-choice questions automatically.	S: timer, auto-submit, auto-grading; M: score equals answer key; T: result within 5 seconds of submission
FREQ-16	US-10	The system shall let a student view only their own grades and attendance for the current and previous terms.	S: own data, two terms; M: no other student's data is reachable; T: available 24/7

FREQ-17	US-11	The system shall send a push notification to a parent when their child is marked absent.	S: push on absent status; M: notification received on the parent's phone; T: within 2 minutes of marking
FREQ-18	US-11	If a push notification cannot be delivered, the system shall send the same alert by SMS.	S: SMS fallback; M: SMS delivered; T: within 5 minutes of the failed push
FREQ-19	US-12	The system shall let a parent pay fees online by Mada card, credit card or Apple Pay through a payment gateway and then issue a receipt.	S: three payment methods plus receipt; M: payment status and receipt saved; T: receipt within 10 seconds of approval
FREQ-20	US-12	The system shall show a parent the outstanding fees, due dates and payment history of each child.	S: three items per child; M: balances equal finance records; T: updated within 1 minute of a payment

FREQ-21	US-13	The system shall show a parent the grades and attendance of each linked child.	S: grades and attendance per child; M: equals the teacher's entries; T: within 1 minute of entry
FREQ-22	US-13	The system shall link a parent account to a child only after an administrator verifies the relationship.	S: administrator verification; M: unlinked children are not visible; T: link active within 1 working day of the request
FREQ-23	US-14	The system shall provide the whole interface in Arabic (right-to-left) and English and remember each user's choice.	S: two languages, saved choice; M: every screen available in both; T: switch takes effect in 1 tap, on the same page
FREQ-24	US-15	The system shall let a parent send a message to a child's teacher and show the teacher's reply in the same thread.	S: one thread per conversation; M: reply appears in the thread; T: message delivered within 1 minute

FREQ-25	US-16	The system shall let an administrator create a student record with name, national ID, date of birth, grade and guardian contact, and reject records with missing fields.	S: five required fields; M: incomplete record cannot be saved; T: validated on save
FREQ-26	US-16	The system shall record the user, time and change for every edit or archiving of a student record.	S: three logged items; M: each edit has a log entry; T: at the moment of the change
FREQ-27	US-17	The system shall warn the administrator when a teacher, room or class is booked twice in the same period.	S: three clash types; M: every double booking is flagged; T: within 2 seconds, before saving
FREQ-28	US-17	The system shall show the published timetable to students, teachers and parents.	S: three viewer groups; M: published version visible to all; T: within 1 minute of publishing

FREQ-29	US-18	The system shall generate attendance and performance reports by class, grade or term in PDF and Excel.	S: three filters, two formats; M: report totals equal stored data; T: within 10 seconds for up to 1,000 students
FREQ-30	US-19	The system shall list each student's outstanding balance, filterable by grade and exportable to Excel.	S: balance list, grade filter, export; M: totals equal finance records; T: updated within 1 minute of a payment
FREQ-31	US-20	The system shall let IT support create and disable accounts and assign one of six roles (principal, teacher, student, parent, administrator, IT).	S: six roles; M: permissions match the role; T: a disabled account is blocked within 1 minute
FREQ-32	US-20	The system shall require multi-factor authentication for every staff account at login.	S: all staff accounts; M: login without the second factor fails; T: on every login

FREQ-33	US-21	The system shall back up all data automatically every day and keep the last 30 backups.	S: daily backup, 30 copies; M: backup log shows success; T: every day at 2:00 AM
FREQ-34	US-21	The system shall let IT support restore the system from a backup.	S: full restore from any of the 30 backups; M: restored data equals the backup; T: completed within 4 hours
FREQ-35	US-22	The system shall send a parent a reminder for each unpaid fee.	S: reminder per unpaid fee; M: notification delivered; T: 7 days and 1 day before the due date
FREQ-36	US-23	The system shall log every view or edit of a student record (user, record, action, time) and let only the principal and IT support read the log.	S: four logged fields, two allowed readers; M: every access has an entry; T: kept for at least 1 year

Exercise 4.3: Review the requirements to be SMART.

Done with part 4.2

Part 6: Key Takeaways

Why Transform User Stories to Requirements?

5 Clarity: User stories are informal and conversational; requirements are formal and precise
6 Implementation Guidance: Requirements provide specific details for developers
7 Testing: Requirements can be tested; user stories are too vague for testing
8 Traceability: Requirements can be tracked and verified
9 Completeness: Transformation often reveals missing functionality

SMART Criteria Explained

Criterion	Definition	Example
Specific	Clear and well-defined, not ambiguous	"Apply 10% discount" not "Apply a discount"
Measurable	Can be quantified or verified	"Maximum 5 reservations" (countable)
Achievable	Realistic and feasible within constraints	Consider technical and business feasibility
Relevant	Aligned with business goals and user needs	Supports the Movie Shop's business model
Time-bound	Specifies when or how often	"At point of sale," "Within 1 second," "Daily"

Common Mistakes to Avoid

10 Too Vague: "The system should be user-friendly" → Too subjective
11 Too Technical: Jumping to implementation details instead of requirements
12 Missing Context: Not considering who, what, and why
13 Incomplete: Forgetting edge cases and error handling
14 Unmeasurable: Requirements that cannot be tested or verified

Submissions :

1- **Lab 3 doc**
2- **Project Features, including Lab 2 Personas and user stories.**